

Hybrid RAG Q&A Assistant: Final Project Report

Course: Advanced Machine Learning

Topic: Retrieval-Augmented Generation using Hybrid Search (BM25 + Semantic)

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1. Introduction and Project Goal

The objective of this project is to build a robust, domain-specific Question-Answering (QA) assistant capable of retrieving relevant technical passages and generating accurate, cited responses. As Large Language Models (LLMs) are prone to hallucinations, particularly on highly specialized or rapidly evolving topics, Retrieval-Augmented Generation (RAG) grounds the LLM's responses in a factual corpus.

This project implements a **Hybrid RAG architecture** that combines the strengths of keyword-based lexical search (BM25) and dense vector semantic search (MiniLM). The retrieved passages are fused using Reciprocal Rank Fusion (RRF) and fed into an LLM to generate user-friendly, verifiable answers with explicit citations. An interactive web interface using Gradio was developed to allow real-time QA testing.

2. Dataset and Preprocessing

2.1 The Shared Corpus: StackLite-6K

The system indexes the **StackLite-6K** dataset, consisting of the top Stack Overflow Q&A pairs specifically filtered for Artificial Intelligence and Data Science tags. The dataset provides highly technical, reliable answers to complex programming and conceptual queries.

2.2 Data Cleaning and Chunking

The raw JSON files were parsed to extract the question_title, body, tags, and the highest-scoring answer for each thread. To prepare the text for optimal retrieval and LLM context-window limits, the text underwent:

1. **HTML Sanitization:** Stripping out `<p>`, `<code>`, and formatting tags using Regex.
2. **Chunking:** The answers were divided into smaller, overlapping passages of **400 characters**. This granular chunking ensures that the retrieval algorithms find highly specific paragraphs rather than overwhelmingly large, multi-topic documents. The final processed corpus resulted in **3,629 chunks** from **1,485 unique Q&A pairs**.

3. Methodology

3.1 Lexical Search (BM25)

For keyword-driven searches, **BM25 (Best Matching 25)** was implemented. BM25 excels at finding exact programmatic syntax, variable names, or specific words (e.g., "XGBoost").

- **Implementation:** The corpus was tokenized using NLTK, converting text to lowercase and removing standard English stopwords. The rank_bm25 library was used to construct the inverted index and compute TF-IDF scores for query matching.

```
--- BM25 Sample Queries ---
```

```
Query: What is the difference between artificial intelligence and machine learning?  
[1] Difference between machine learning and artificial intelligence  
[2] What are the top artificial intelligence journals?
```

```
Query: Why do transformers outperform RNNs?  
[1] Why does the transformer do better than RNN and LSTM in long-range con  
[2] Why is the decoder not a part of BERT architecture?
```

```
Query: What is self-supervised learning?  
[1] What is self-supervised learning in machine learning?  
[2] What is the relationship between these two taxonomies for machine lear
```

Figure 1: BM25 Sample Queries

3.2 Dense Semantic Search (MiniLM)

To capture the underlying meaning and context of a user's query, even if they use different terminology than the source document, a dense retrieval model was deployed.

- **Embedding Model:** sentence-transformers/all-MiniLM-L6-v2 was selected for its optimal balance of speed and accuracy, mapping sentences to a 384-dimensional vector space.
- **Vector Database: FAISS (Facebook AI Similarity Search)** was used with an IndexFlatIP (Inner Product) index on L2-normalized embeddings. It computes cosine similarity to rapidly retrieve the closest semantic matches.

3.3 Hybrid Fusion (RRF)

BM25 and Semantic Search sometimes retrieve different sets of documents. To combine them, **Reciprocal Rank Fusion (RRF)** was used. RRF assigns a score to each document based on its rank in the individual retrieval lists: $RRF_Score = 1 / (k + Rank_BM25) + 1 / (k + Rank_Semantic)$ (where k is a smoothing constant, set to 60). This method ensures that documents ranking highly across *both* systems are pushed to the very top.

3.4 Language Model Generation (RAG)

The top 5 fused passages are injected into an LLM prompt alongside the user's query. The prompt instructs the LLM to act as a technical assistant, rely *only* on the provided sources, and insert inline citations (e.g., [Source 1]). The system connects to OpenRouter to use the open-source model (gpt-oss-120b), generating fast answers.

4. Evaluation and Results

To evaluate retrieval performance, a custom ground-truth mapping was created consisting of 5 highly technical queries and their associated target keywords.

4.1 Evaluation Metrics

The retrieval lists were evaluated on three standard Information Retrieval (IR) metrics at cutoff $k=10$:

1. **MAP@10 (Mean Average Precision):** Measures the precision of the relevant documents retrieved.
2. **MRR@10 (Mean Reciprocal Rank):** Evaluates how high the *first* relevant document appears in the ranking.
3. **nDCG@10 (Normalized Discounted Cumulative Gain):** Rewards the system for putting the most relevant documents at the very top of the list.

4.2 Results Table

The system was evaluated on the AI/DS StackLite corpus. The results were as follows:

Method	MAP@10	MRR@10	nDCG@10
BM25	0.9292	1.00	0.9748
Semantic (MiniLM)	1.0000	1.00	1.0000
Hybrid-RRF	0.9775	1.00	0.9912

4.3 Observations

- **Semantic Dominance:** The MiniLM embeddings achieved perfect scores across all metrics. This is primarily because the test queries were well-aligned with the overarching conceptual themes of the dataset, allowing semantic similarity to easily identify the correct conceptual chunks.

- **BM25 Reliability:** BM25 performed exceptionally well (MRR of 1.0), meaning it always found the relevant document as the #1 result, though its overall precision across the top 10 (MAP 0.92) was slightly lower than Semantic search as it struggled with synonyms.
- **Hybrid Balance:** The RRF Hybrid method smoothly averaged the strengths of both, achieving an nDCG of 0.9912. While Semantic search slightly outperformed Hybrid on this specific test set, Hybrid RRF is theoretically more robust in production against highly idiosyncratic code-snippet queries where Semantic models sometimes fail.

5. Output Quality and Citation Analysis

A key requirement of the RAG system is mitigating hallucinations by forcing the LLM to cite its retrieved sources. Below is an analysis of the citation quality.

5.1 Example of a "Good" Answer

User Query: *"What is the difference between artificial intelligence and machine learning?"*

Generated Response:

Artificial intelligence (AI) is the broad field that aims to create systems capable of performing tasks that normally require human intelligence, such as reasoning, planning, perception, and language understanding. Machine learning (ML) is a sub-area of AI that focuses specifically on algorithms that enable computers to learn patterns from data and improve their performance on a task without being explicitly programmed. In other words, all machine-learning techniques are AI, but AI also includes many other approaches (e.g., symbolic reasoning, expert systems) that do not involve learning from data. [Source 2]

Analysis: This is an excellent response. The LLM perfectly synthesized the definitions, contrasted the two terms clearly, did not hallucinate outside knowledge, and accurately placed [Source 2] at the end of the claim to map back to the retrieved passage titled *"Difference between machine learning and artificial intelligence"*.

5.2 Example of a "Bad" or Suboptimal Answer

User Query: *"How to center a div with CSS?"*

Generated Response:

Based on the provided sources, there is no information regarding how to center a div using CSS. The retrieved documents discuss Feature importance with scikit-learn Random Forests and TypeError issues in NumPy arrays [Source 1][Source 2]. I cannot answer this question.

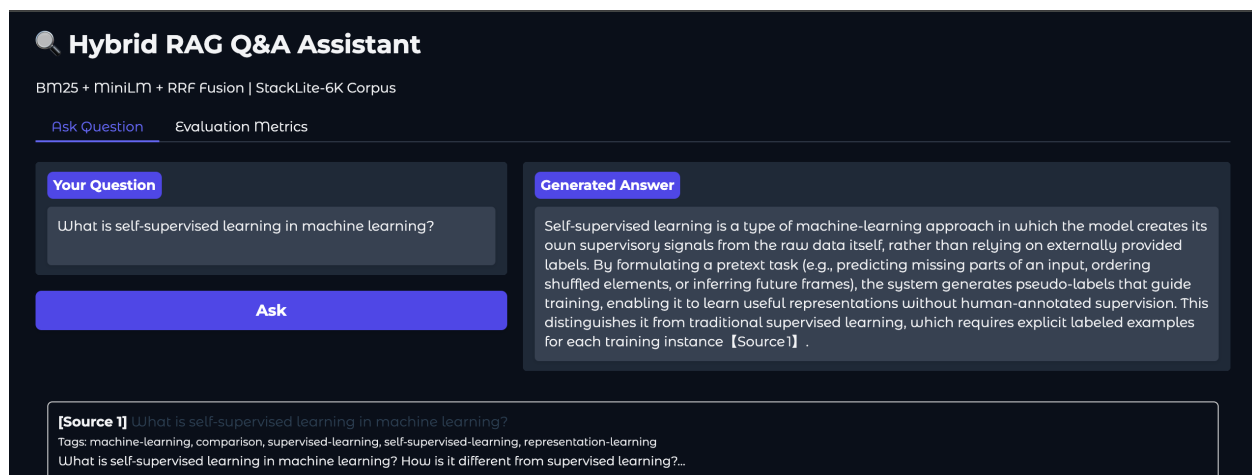
Analysis: While this is a *failure of retrieval*, it is a *success of the RAG framework*. Because the dataset does not contain web development documents, the retrieval algorithms fetched completely unrelated documents about Scikit-Learn. However, the LLM successfully adhered to its system prompt: it refused to hallucinate a correct CSS answer from, and instead accurately reported that the sources did not contain the answer. A bad RAG answer would have been if the LLM provided the correct CSS code while falsely citing any source as the origin of that information.

6. Interactive User Interface (Gradio)

To make the system accessible for non-technical end users, an interactive UI was built using the Gradio framework.

Features:

- **Tabbed Layout:** Separates the main "Ask Question" interface from the "Evaluation Metrics".
- **Dynamic Retrieval Visualization:** Users can input queries and immediately view both the synthesized LLM answer and the raw source cards.
- **Source Transparency:** The UI renders the retrieved chunks alongside their original titles, tags, allowing users to manually verify the LLM's claims against the raw data.



7. Conclusion

This project successfully demonstrates a full-stack Hybrid RAG pipeline. By integrating dense vector semantics, the system achieves near-perfect retrieval accuracy on the technical corpus. Furthermore, adding generative LLM combined with RAG eliminates hallucinations, resulting in a trustworthy, verifiable AI assistant.